```python
import pandas as pd
import numpy as np

file_name = "Sales (1).xlsx"
xls = pd.ExcelFile(file_name)
df = pd.read_excel(xls, sheet_name=xls.sheet_names[0])

df['Month_sales'] = pd.to_datetime(df['Month_sales'])
df['Quarter'] = df['Month_sales'].dt.quarter
quarterly_sales = df.groupby('Quarter')['Total'].sum()

sales_data = quarterly_sales.to_numpy()

total_sales = np.sum(sales_data)

q1_sales = sales_data[0]
q4_sales = sales_data[3]

percentage_increase = ((q4_sales - q1_sales) / q1_sales) * 100

print(f"Sales Data Array: {np.round(sales_data, 2)}")
print(f"Total Sales for the Year: ${total_sales:,.2f}")
print(f"Percentage Increase (Q1 to Q4): {percentage_increase:.2f}%")
```

```
Sales Data Array: [1266978.5  1246529.71 1185251.76 1320505.26]
Total Sales for the Year: $5,019,265.23
Percentage Increase (Q1 to Q4): 4.22%
```

```python
import numpy as np

sales_data = np.array([
    [120, 125, 130],
    [50, 55, 60],
    [200, 210, 220]
])

overall_average_price = np.mean(sales_data)
print(f"Overall Average Price: ${overall_average_price:.2f}")

average_per_product = np.mean(sales_data, axis=1)

print("\nAverage Price per Product:")
for i, avg in enumerate(average_per_product, start=1):
    print(f"Product {i}: ${avg:.2f}")
```

```
Overall Average Price: $130.00

Average Price per Product:
Product 1: $125.00
Product 2: $55.00
Product 3: $210.00
```

```python
import pandas as pd
import numpy as np

df = pd.read_csv("house_data_large.csv")
house_data = df.to_numpy()

bedroom_column_index = 0
sale_price_column_index = 2

more_than_four_bedrooms_mask = house_data[:, bedroom_column_index] > 4

filtered_houses = house_data[more_than_four_bedrooms_mask]
average_sale_price = np.mean(filtered_houses[:, sale_price_column_index])

print(f"Average sale price of houses with >4 bedrooms: ${average_sale_price:,.2f}")
```

```
Average sale price of houses with >4 bedrooms: $612,153.85
```

```python
import pandas as pd
import numpy as np
file_name = 'student_scores - student_scores (2) (1).csv'
df = pd.read_csv(file_name)
subjects = ['Math', 'Science', 'English', 'History']
student_scores = df[subjects].to_numpy()
average_scores = np.mean(student_scores, axis=0)
print("Average Scores per Subject:")
for subject, avg in zip(subjects, average_scores):
    print(f"{subject}: {avg:.2f}")
highest_avg_index = np.argmax(average_scores)
highest_subject = subjects[highest_avg_index]
highest_score = average_scores[highest_avg_index]

print(f"\nThe subject with the highest average score is {highest_subject} ({highest_score:.2f}).")
```

```
Average Scores per Subject:
Math: 81.97
Science: 82.56
English: 82.97
History: 82.88

The subject with the highest average score is English (82.97).
```

```python
import pandas as pd
import numpy as np

file_name = "Fuel_Efficiency datasets.xlsx - Sheet1 (1).csv"
df = pd.read_csv(file_name)

fuel_efficiency = df['highway_mpg(Fuel Efficiency)'].to_numpy()

average_fuel_efficiency = np.mean(fuel_efficiency)

model_A_mpg = fuel_efficiency[0]
model_B_mpg = fuel_efficiency[1]

percentage_improvement = ((model_B_mpg - model_A_mpg) / model_A_mpg) * 100

print(f"Average Fuel Efficiency: {average_fuel_efficiency:.2f} MPG")
print(f"Model A MPG: {model_A_mpg}")
print(f"Model B MPG: {model_B_mpg}")
print(f"Percentage Improvement (Model A to Model B): {percentage_improvement:.2f}%")
```

```
Average Fuel Efficiency: 28.61 MPG
Model A MPG: 36
Model B MPG: 37
Percentage Improvement (Model A to Model B): 2.78%
```